

PO-BACKED INTERNATIONAL CASE STUDY

FROM CONSULTING TO COMMISSIONING

A complete industrial technology foundation—designed, supplied, implemented and documented by one accountable solution partner.

CONSULT

DESIGN

SOLUTION

IMPLEMENT

KRYSTAL INGREDIENTS PRIVATE LIMITED • DAHEJ

PO D/FA/25-26/13089 • 22 MAY 2025

PO D/FA/25-26/13090 • 27 MAY 2025

Prepared by D-TECH Solution Integrators Pvt. Ltd.

NAVIGATION

Complete case-study navigation

Each section connects the commercial PO scope to the consulting, design, implementation and commissioning story.

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SOURCE CONTROL

Quantities and models are reproduced from PO 13089 and PO 13090.

EXECUTIVE OVERVIEW

One integrated industrial platform

D-TECH converted two purchase orders into one coordinated operating environment spanning infrastructure, security, connectivity and collaboration.



DELIVERY MANDATE

Design and implement the complete digital backbone—not isolated product installations.

- Central compute, storage and identity foundation
- FortiGate-protected edge with three-year UTM
- Managed fibre, PoE switching and Wi-Fi 6
- 167-camera surveillance with dual 128-channel NVRs
- Access control, video collaboration and user endpoints

167

CAMERAS

20

MANAGED SWITCHES

8

WI-FI 6 APS

1,500m

SM OFC

01 Consult

Convert operational and facility requirements into an approved design basis.

02 Design

Create integrated topology, capacity, coverage and service boundaries.

03 Implement

Supply, install, configure, test, document and hand over.

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Network • CCTV • Wi-Fi • Access • VC

PO 13090 • 27 MAY 2025

Server • Storage • Firewall • Endpoints

COMMERCIAL FOUNDATION

Two POs. One coordinated solution.

The purchase orders define complementary technology layers that D-TECH integrated into a common architecture.

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NETWORK, SECURITY & COLLABORATION

- 167 IP cameras + dual 128-channel NVR
- 20 managed switches + 28 optics
- 8 Wi-Fi 6 access points
- 1,500 m single-mode armoured OFC
- Two access-controlled doors
- Two Logitech video-conference systems
- Racks, LIUs, patching and CAT6

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COMPUTE, STORAGE & SECURITY EDGE

- Lenovo ThinkSystem SR650 V2 server
- QNAP TS-464 network storage
- FortiGate 70F security appliance
- Three-year FortiGate UTM protection
- Five ASUS ExpertCenter desktops
- Five ASUS ExpertBook laptops
- Installation and configuration services

01 PO validation

Models, quantities and service lines reviewed against the client purchase orders.

02 Design consolidation

Both orders mapped into one logical and physical architecture.

03 Closure evidence

Product register and completion certificate aligned to the PO scope.

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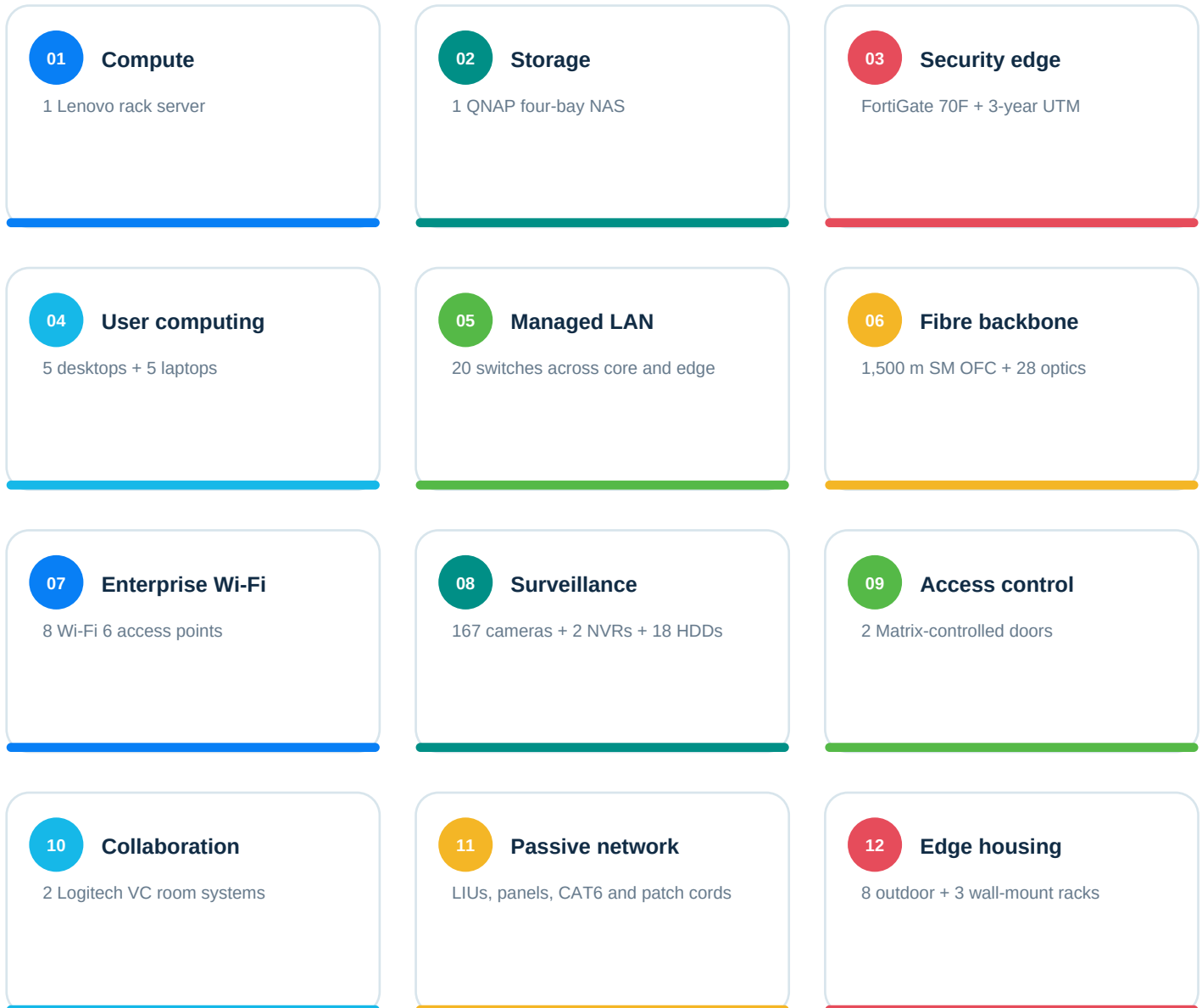
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COMPLETE SOLUTION SCOPE

Every technology layer—engineered as one

The delivery scope combines active infrastructure, passive pathways, security, collaboration and endpoint computing.



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CONSULTANT • DESIGN

Decisions before devices

D-TECH acted as consultant, solution architect, system integrator and implementation partner across both POs.

01

Discover

Operational zones, users, surveillance, voice and business-service requirements.

02

Survey

Rack positions, fibre routes, cable pathways, coverage and power/interface constraints.

03

Engineer

Topology, capacities, uplinks, recording, security boundaries and room solutions.

04

Integrate

Product selections from both POs mapped into one governed architecture.

05

Validate

Connectivity, functionality, recording, access, Wi-Fi, voice and handover evidence.

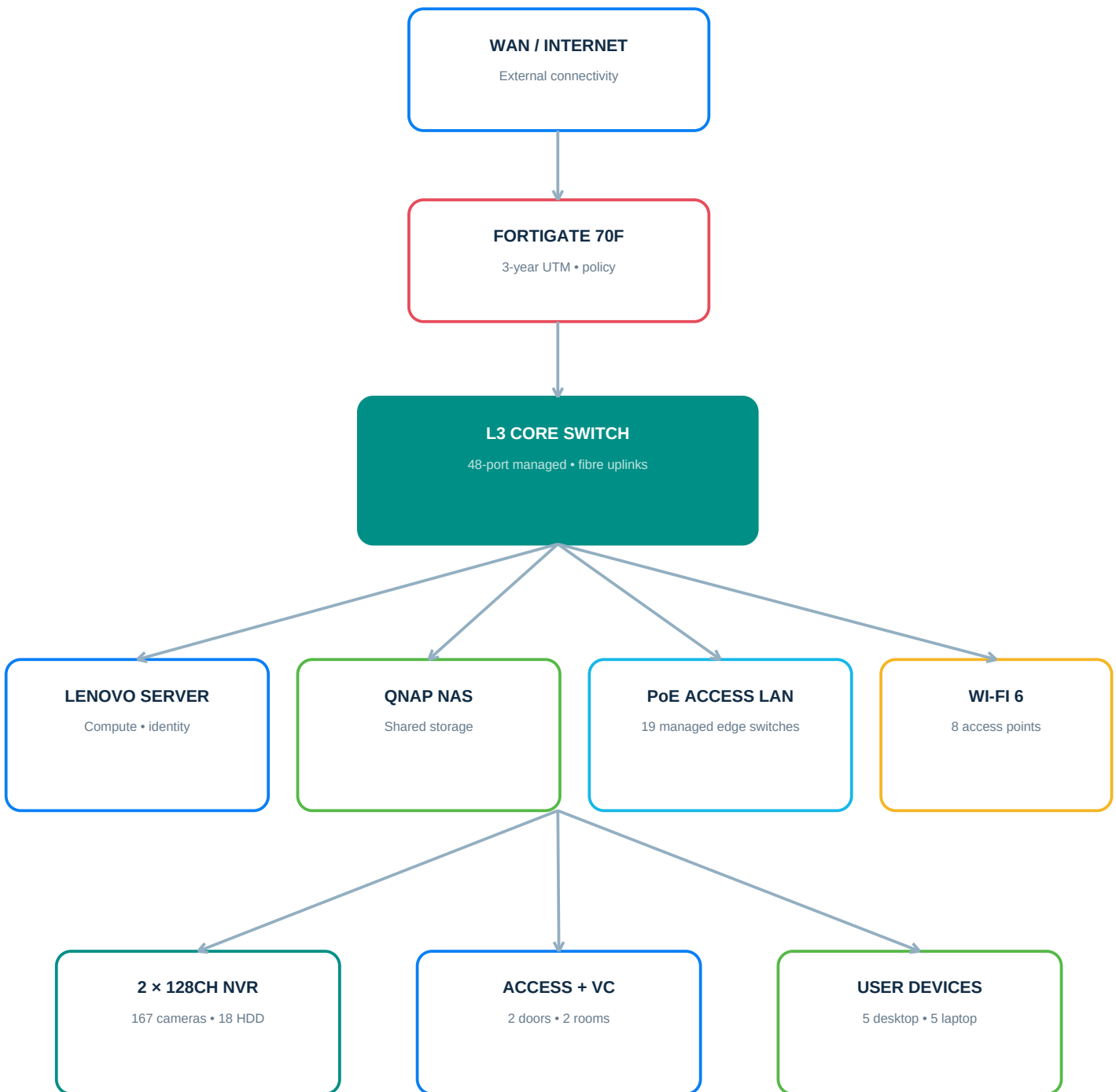
CONSULTANT PRINCIPLE

The right solution is not a product list. It is an operational design in which every product has a role, a connection, an owner and a test method.

CANVAS SOLUTION DRAWING

Secure edge. Resilient core. Connected services.

The consolidated architecture shows how PO 13090 compute and security connect to the PO 13089 network, surveillance and collaboration layers.



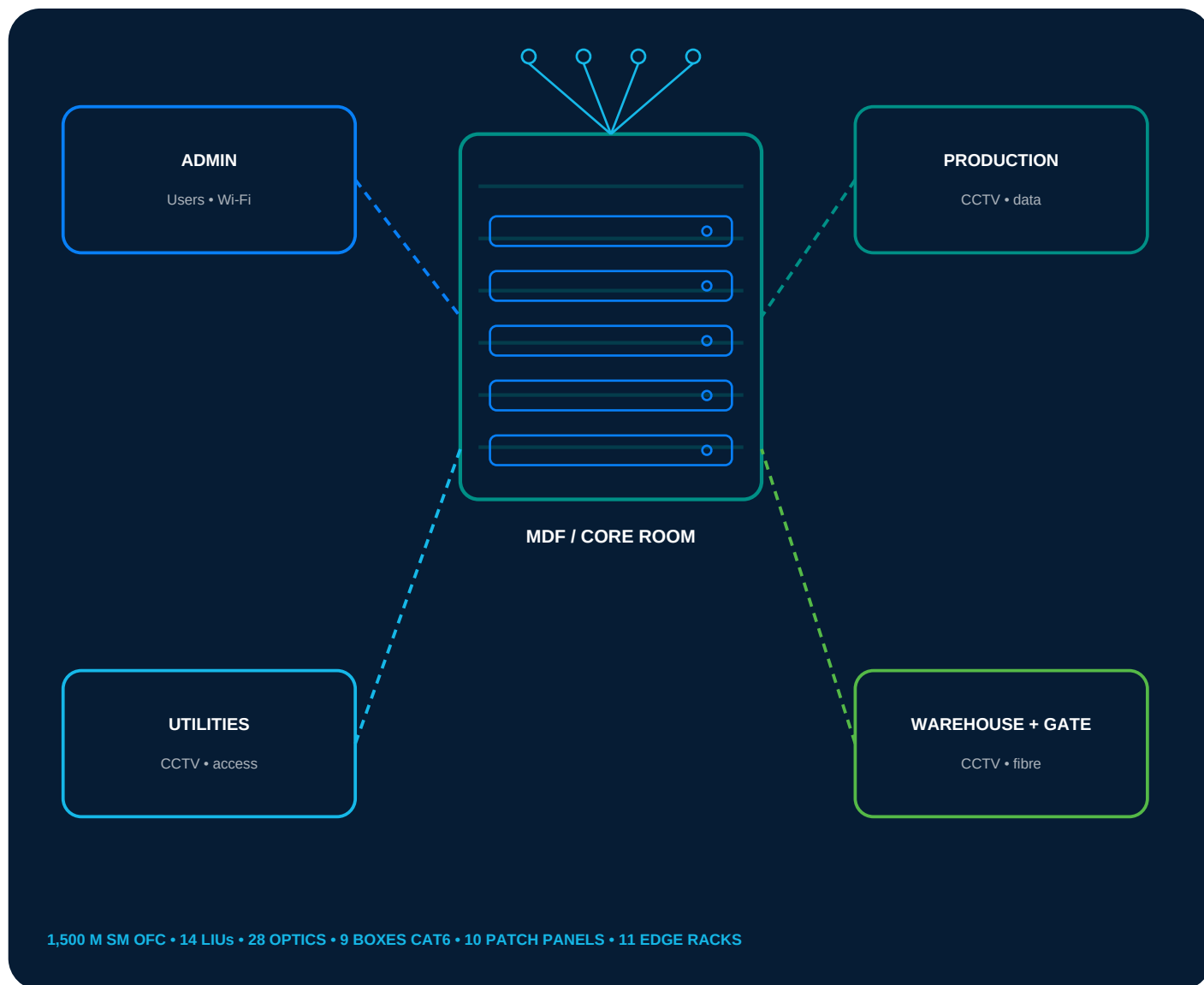
DESIGN INTENT

Central control, segregated services, PoE distribution, fibre resilience and documented growth capacity.

CANVAS SITE SKETCH

From the core room to every operating zone

The physical design combines single-mode fibre, managed distribution, CAT6 access and weather-appropriate edge racks.



01 Backbone

Armoured single-mode fibre connects the distributed zones.

02 Distribution

Managed PoE switching supports cameras, APs and users.

03 Serviceability

Racks, panels, LIUs and patching create traceable pathways.

SECURITY BY DESIGN

Protection across edge, identity and physical access

The solution combines network security, managed access and controlled door entry rather than relying on one perimeter device.

01

FORTIGATE EDGE

FortiGate 70F appliance with three-year UTM protection.

02

SERVICE SEGMENTATION

Separate functional domains for server, storage, users, CCTV, Wi-Fi and collaboration.

03

MANAGED ACCESS

Layer-3 core, PoE distribution and documented fibre uplinks.

04

PHYSICAL ACCESS

Two Matrix COSEC PATH controllers with EM locking assemblies.

05

OPERATIONAL EVIDENCE

PO register, topology, device schedules and acceptance records.

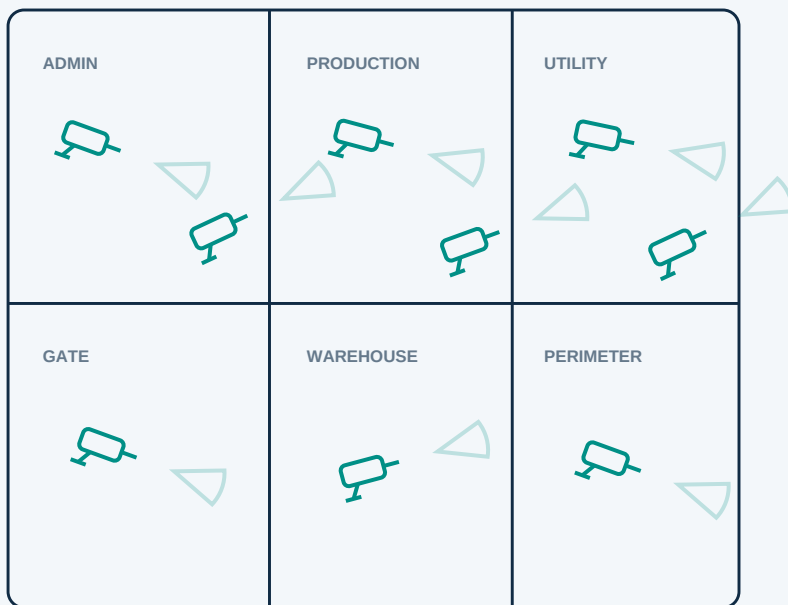
SECURITY OUTCOME

Technology, identity, traffic and physical entry are governed through defined controls and documented ownership.

INDUSTRIAL SURVEILLANCE

167 cameras. Dual NVR architecture.

PO 13089 provides comprehensive indoor/outdoor monitoring capacity with recording, storage and pole-mounted coverage.



CONCEPTUAL COVERAGE DRAWING • NOT TO SCALE

61

2MP TURRET CAMERAS

Prama PT-NC232PI-WNM(D2)

106

4MP BULLET CAMERAS

Prama PT-NC340PI-N(D2)

2 × 128CH

RECORDING CAPACITY

Prama PT-NR3A128-16I

01 Storage

18 WD surveillance drives support the NVR recording layer.

02 Coverage

13 GI poles extend visibility across external areas.

03 Operations

Live view, recording, playback and controlled evidence retrieval.

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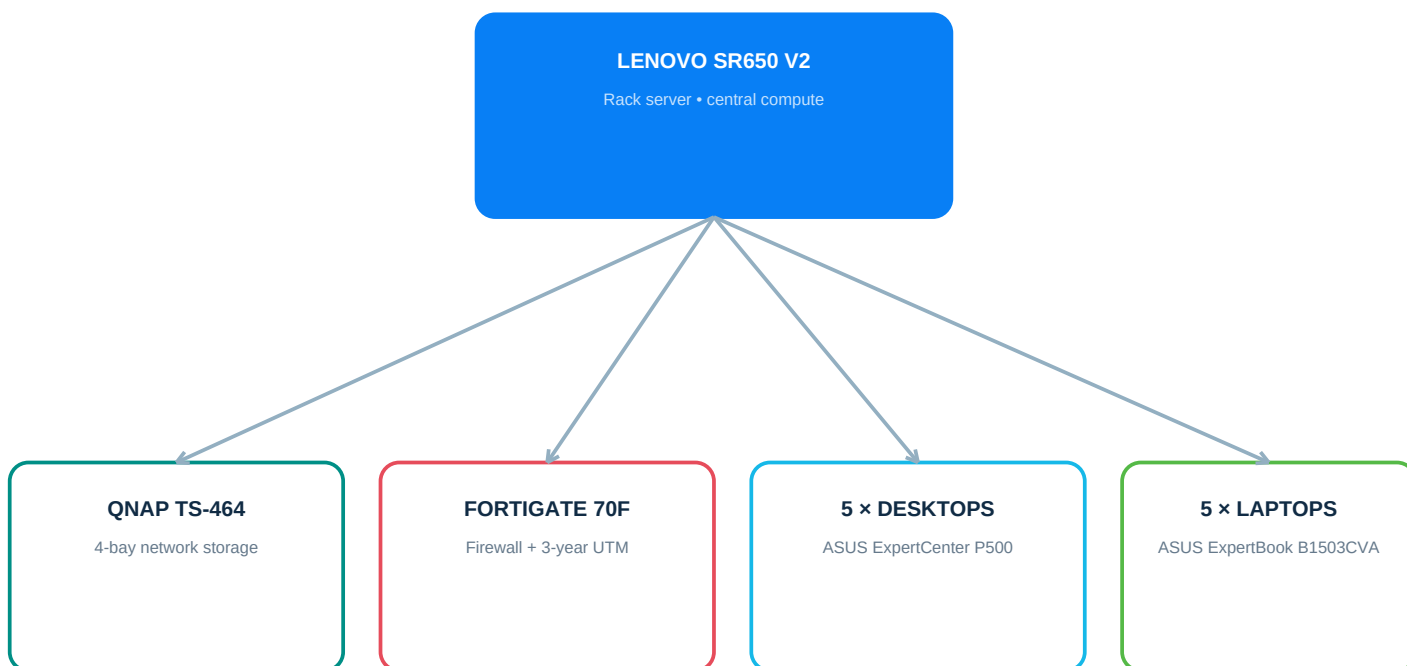
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COMPUTE • STORAGE • ENDPOINTS

A secure foundation for users and services

PO 13090 establishes the compute, storage, protected edge and business endpoint layer.



IMPLEMENTATION RESPONSIBILITY

- Rack and base installation
- Server configuration
- NAS setup and access
- Firewall deployment
- UTM activation
- Endpoint readiness

1

SERVER

Lenovo ThinkSystem SR650 V2

2

STORAGE + FIREWALL

QNAP NAS + FortiGate

10

USER DEVICES

5 desktops + 5 laptops

IMPLEMENTATION AND EVIDENCE

From purchase order to operational handover

Every stage connects commercial scope, engineering intent, installation quality and client acceptance.

1

Validate PO

Confirm models, quantities and service responsibilities.

2

Survey and design

Finalise locations, routes, capacities and interfaces.

3

Supply and stage

Inspect assets, prepare baselines and sequence delivery.

4

Install and configure

Build racks, fibre, LAN, compute, CCTV, access and collaboration.

5

Integrate and test

Verify connectivity, security, recording, access and user services.

6

Document and hand over

Issue product register, as-built records and completion certificate.

COMMISSIONING EVIDENCE

PO schedules • topology • rack and cable checks • connectivity tests • NVR recording/playback • access control • Wi-Fi • firewall • compute/storage • collaboration • acceptance certificate

PRODUCT REGISTER A

PO 13089 - surveillance, access and collaboration

Consolidated product schedule from D/FA/25-26/13089 dated 22 May 2025.

QTY	PRODUCT / MODEL	FUNCTION
2	Logitech Group VC systems Group room kits	Video collaboration
2	Expansion microphones Logitech Group expansion mic	Extended room pickup
2	10 m expansion cables Logitech Group cable	Room extension
2	128-channel NVR Prama PT-NR3A128-16I	Central recording
61	2MP fixed turret cameras Prama PT-NC232PI-WNM(D2)	Indoor monitoring
106	4MP smart dual-light bullet cameras Prama PT-NC340PI-N(D2)	Outdoor monitoring
18	Surveillance hard disks WD surveillance class	Recording storage
13	GI camera poles External mounting poles	Perimeter coverage
2	Access controllers Matrix COSEC PATH DCFE	Door access control
2	Duct plates Matrix COSEC PATH duct plate	Controller mounting
2	EM locks 600 lbs single-leaf 12V DC	Door locking
2 sets	Lock brackets L bracket + door-lock bracket	Door hardware
Included	Installation and configuration VC, NVR, cameras and access	Commissioning services

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PRODUCT REGISTER B

PO 13089 - active and passive network

Managed switching, Wi-Fi 6, fibre backbone, structured cabling and edge housing.

QTY	PRODUCT / MODEL	FUNCTION
1	48-port managed Layer-3 switch Digisol DG-GS4652SE2	Core switching
8	24-port managed PoE+ switches Digisol DG-GS1528HP/C	Distribution / access
11	8-port smart managed PoE+ switches Digisol DG-GS1508PL	Edge PoE access
2	10G SFP+ transceivers Digisol DG-SA2143	High-speed uplink
26	1G single-mode SFP transceivers Digisol DG-SA1133-20	Fibre uplinks
8	Wi-Fi 6 access points Netgear WAX220 • 2.5G PoE	Enterprise wireless
1,500 m	6-core SM armoured OFC D-Link NCE-FS09/0-AUHD-06	Fibre backbone
14	12-port loaded LIUs D-Link LC single-mode	Fibre termination
26	LC-LC SM duplex patch cords 3 metre	Optical patching
9 boxes	CAT6 UTP cable D-Link 305 m box • grey	Horizontal cabling
7	48-port loaded patch panels D-Link NPP-C61BLK48L1	Copper termination
3	24-port loaded patch panels D-Link NPP-C61BLK24L1	Copper termination
8	4U outdoor network racks Elixir 600W × 450D	Outdoor edge housing
3	6U wall-mount racks Valrack 600W × 600D	Indoor edge housing

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PRODUCT REGISTER C

PO 13090 - compute, storage and firewall

Consolidated product schedule from D/FA/25-26/13090 dated 27 May 2025.

QTY	PRODUCT / MODEL	FUNCTION
1	Rack server Lenovo ThinkSystem SR650 V2	Central compute
5	Mini-tower desktops ASUS ExpertCenter P500 • D500MD	Business endpoints
5	Business laptops ASUS ExpertBook B1503CVA	Mobile endpoints
1	4-bay network storage QNAP TS-464	Central NAS storage
1	Firewall appliance FortiGate 70F	Secure internet edge
1	Three-year UTM protection FortiGate 70F subscription	Threat protection
Included	Server installation/configuration Dahej site service	Commissioning
Included	Storage installation/configuration Dahej site service	Commissioning
Included	Firewall installation/configuration Dahej site service	Commissioning

REGISTER CONTROL

The descriptions above are consolidated for case-study clarity. The signed purchase orders remain the commercial authority for exact line codes, rates, tax and terms.

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OUTCOME AND PARTNERSHIP

A complete foundation—with PO-level traceability

The result is a unified platform where products, design decisions, installation activities and acceptance evidence connect through one accountable delivery partner.

DESIGNED AS ONE SYSTEM

Server, storage, firewall, switching, fibre, Wi-Fi, CCTV, access control, collaboration and endpoints now form one documented technology foundation.

01 Consultant-led

Requirements connected to operational context.

02 Designed

Architecture and capacities defined before installation.

03 Implemented

Two PO scopes integrated into one environment.

04 Traceable

Product register and acceptance documents aligned.

D-TECH DELIVERY PROMISE

Consult with context. Design with discipline. Implement with evidence. Support with accountability.